

An atom at zero forces zero exponential persistence rate

Literature-implied disproof of Remark 3 in arXiv:2112.04820

Automated Mathematics Run `fa_banach_001`, lane 2

19 July 2026

Status. Literature-implied answer: a separate 2026 theorem yields a full counterexample to the 2021 conjecture after a direct one-dimensional specialization. This is not claimed as a new mathematical result.

1 Source conjecture

Feldheim, Feldheim, and Mukherjee [1] define

$$\rho'(0) = \lim_{\varepsilon \downarrow 0} \frac{\rho([- \varepsilon, \varepsilon])}{2\varepsilon}, \quad \mathcal{M} = \{\rho : \rho'(0) \in (0, \infty]\},$$

within their finite log-moment class $\mathcal{L}$. Their Remark 3, PDF page 3, conjectures that for every $\ell > 0$ and $\rho \in \mathcal{M}$, the exponential persistence exponent

$$\theta_\rho^\ell = - \lim_{T \rightarrow \infty} \frac{1}{T} \log \mathbb{P} \left(\inf_{t \in [0, T]} f_\rho(t) > \ell \right)$$

is always positive.

Remark 3. For $\ell \leq 0$ and $\rho \in \mathcal{M}$, it follows from our results that θ_ρ^ℓ is positive if and only if $\rho_{ac} \neq 0$. For $\ell > 0$ and $\rho \in \mathcal{M}$ we conjecture that θ_ρ^ℓ is always positive.

Figure 1: The complete conjecture in Remark 3 of the source paper [1].

2 Supporting result and identification

Feldheim, Feldheim, and Muirhead [2] later consider stationary Gaussian fields on $\mathbb{R}^d$ or $\mathbb{Z}^d$. Their Theorem 3(iv), PDF pages 5–6, applies when the spectral measure has an atom at the origin, a nonzero absolutely continuous part, and no singular-continuous part. Writing $m = \|\rho_{ac}\|$, their formula (1.10), in the case $d = 1$ and $\alpha = 0$, gives

$$-\log \mathbb{P}(f_\rho(t) \geq q \text{ for every } t \in [-T, T]) \sim \frac{m}{\rho(\{0\})} \log T. \tag{1}$$

The paper states immediately before its main theorem that the results extend from level zero to every fixed level $q \in \mathbb{R}$.

Proposition 1. *The conjecture in Remark 3 of [1], as stated, is false. In fact it fails for a nondegenerate process with a smooth nonzero absolutely continuous spectral component.*

Proof. Fix $a > 0$ and set

$$d\rho(\lambda) = a d\delta_0(\lambda) + e^{-\lambda^2} d\lambda.$$

This is a finite symmetric nonnegative spectral measure with finite log-moment, so $\rho \in \mathcal{L}$. Moreover,

$$\rho'(0) = \lim_{\varepsilon \downarrow 0} \frac{a + \int_{-\varepsilon}^{\varepsilon} e^{-\lambda^2} d\lambda}{2\varepsilon} = +\infty,$$

hence $\rho \in \mathcal{M}$. It has no singular-continuous part and has $m = \|\rho_{ac}\| = \sqrt{\pi} > 0$, so the later theorem applies.

Fix the conjecture's level $\ell > 0$ and apply (1) at the higher fixed level $q = \ell + 1$. Stationarity identifies persistence on $[0, T]$ with persistence on $[-T/2, T/2]$ after translation. Consequently

$$\mathbb{P}\left(\inf_{t \in [0, T]} f_\rho(t) > \ell\right) \geq \mathbb{P}(f_\rho(t) \geq \ell + 1 \text{ for all } t \in [-T/2, T/2]) = T^{-\sqrt{\pi}/a + o(1)}.$$

The quantity $-T^{-1} \log$ of the left-hand probability is nonnegative and has limsup at most 0. Its limit therefore exists and equals 0. Thus $\theta_\rho^\ell = 0$, contrary to the conjectured strict positivity. $\square$

3 Status, limitations, and search record

The pure atom $\rho = \delta_0$ gives the still simpler constant-process counterexample: $f_\rho(t) = Z$ and the persistence probability is $\mathbb{P}(Z > \ell)$ for every T . The construction above is stronger because it remains nondegenerate and is directly covered by the later literature.

The run indexes were searched for arXiv:2112.04820 and the core persistence, spectral-measure, and atom-at-zero phrases. Targeted arXiv searches through 19 July 2026 found [2]. That paper cites [1] and proves the decisive asymptotic, but does not explicitly announce that Remark 3 is being disproved. Accordingly this packet is classified as a literature-implied answer, not an original counterexample. Human review should check the fixed level extension stated on PDF page 3 and the specialization of Theorem 3(iv).

References

- [1] N. D. Feldheim, O. N. Feldheim, and S. Mukherjee, *Persistence and Ball Exponents for Gaussian Stationary Processes*, Commun. Pure Appl. Math. 78 (2025), 1949–2000; arXiv:2112.04820.
- [2] N. Feldheim, O. Feldheim, and S. Muirhead, *Persistence and entropic repulsion of stationary Gaussian fields with spectral singularity at the origin*, arXiv:2605.20587 (2026).